Supplementary Material for "From Articles to Publishers: Aggregating Language Model Predictions for News Source Reliability Inference"

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This supplementary material accompanies the main manuscript submitted to IEEE Transactions on Computational Social Systems.

1 Model performance

Figure 1 shows the confusion matrices for each fold and for the average.

2 Publisher-level Results

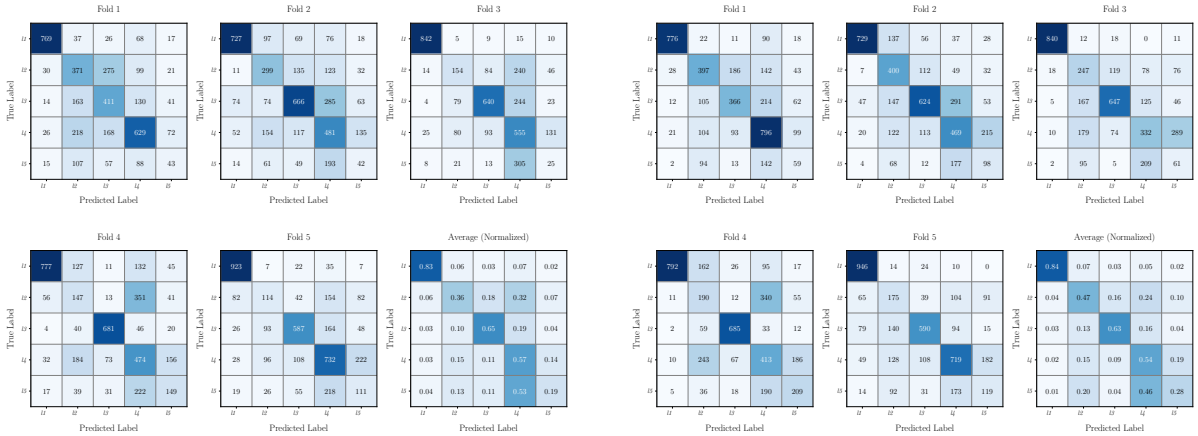
Figure 2 reports the publisher-level confusion matrices obtained after aggregating article-level predictions by majority voting.

The matrices reveal that errors are strictly local. The most frequent confusion occurs between l_4 (Generally Credible) and l_5 (High Credibility). As observed at the article level, distinguishing l_5 from l_4 is inherently difficult because l_5 represents a strict ceiling (score = 100) while l_4 covers the immediately lower high-credibility range (75–99). Misclassifying these two adjacent levels constitutes a minor severity error, as both indicate high reliability. No model exhibits systematic confusion across non-adjacent levels, confirming that aggregation stabilizes predictions.

3 Political leaning

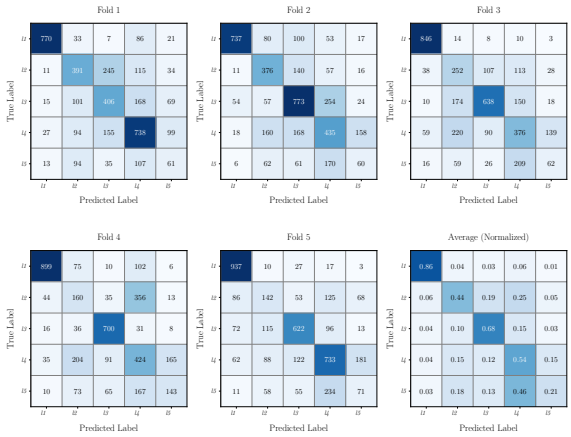
Figure 3 reveals why the Chi-square test rejects the hypothesis of independence: the distribution of errors varies between moderate and extreme political groups, with notable differences across models. The matrices for 'Slightly Right' and 'Non-labelled' orientations exhibit strong diagonal dominance across all architectures. For the former (the largest group, $N = 7490$), Mistral, RoBERTa, and BERT achieve comparable accuracies of 69.0%, 68.4%, and 67.6%, respectively, with Llama 3 at 66.4%. The domain counts (annotated as 'Dom' in the cells) show that this performance is robust across a large number of unique sources.

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[5]This work is partially supported by IIT-CNR internal project SIMULARE 'SIMULAzione delle Reazioni alla disinformazionE con LLM'.

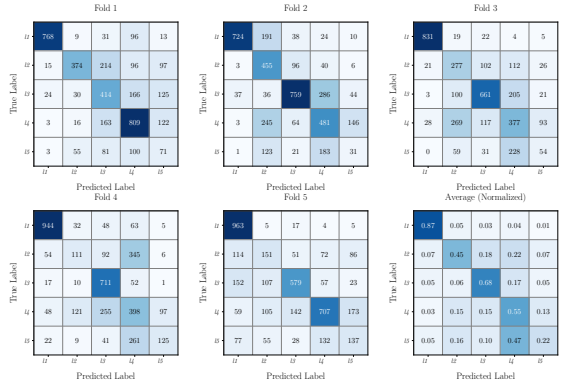


(a) BERT

(b) RoBERTa

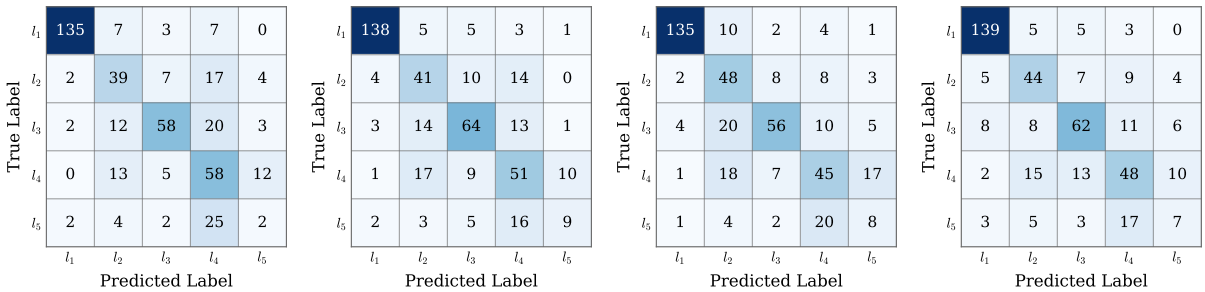


(c) Llama 3



(d) Mistral

Figure 1: Confusion matrices showing the models' performance across five cross-validation folds and their overall average. The rows correspond to the true labels and the columns to the predicted ones.



(a) BERT (b) RoBERTa (c) Llama 3 (d) Mistral

Figure 2: Publisher-level confusion matrices for BERT, RoBERTa, Llama 3, and Mistral. Rows indicate true reliability levels, columns indicate predicted levels.

In contrast, the 'Far Left' ($N = 1015$) and 'Far Right' ($N = 2142$) matrices display high off-diagonal scattering and significantly lower global accuracy. For 'Far Left', all models perform poorly with accuracies between 28.1% and 29.5% for the three smaller models and 36.5% for Mistral. Visual inspection confirms a systematic directional bias: predictions frequently land



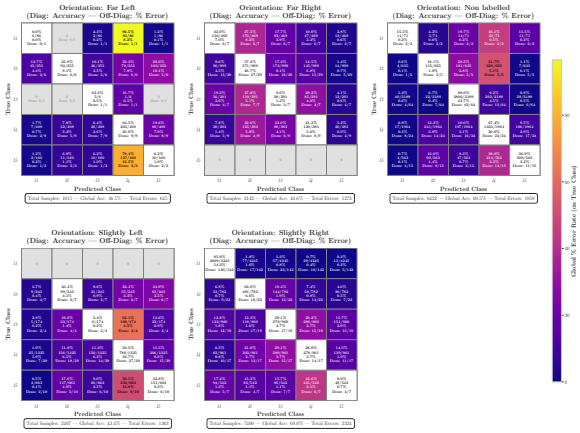
(a) BERT



(b) RoBERTa



(c) Llama 3



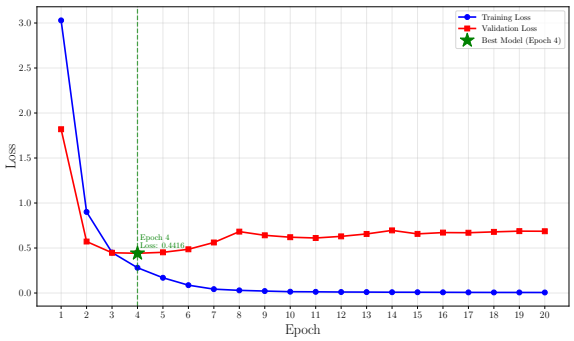
(d) Mistral

Figure 3: Confusion matrices for political orientation across the four models. The rows correspond to the true labels and the columns to the predicted ones. The cells show class-wise percentages and counts, dataset-level percentages and contributing domains, and the colour represents the error rate per true class.

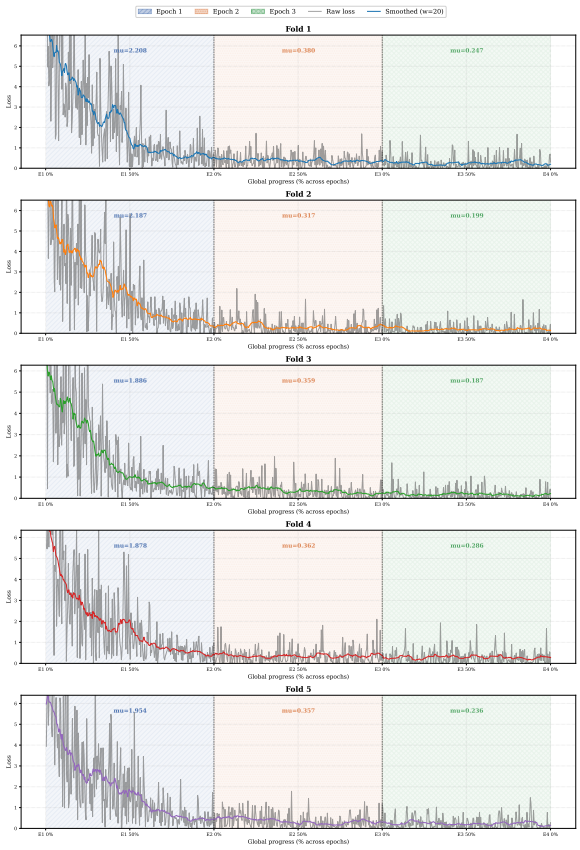
in the upper triangle of the matrix, confirming that models overestimate the reliability of these sources.

The 'Far Right' category shows the greatest difference between the models. Mistral, Llama 3, and RoBERTa achieved 40.6%, 37.8%, and 37.3%, respectively, while BERT dropped to 31.7%.

Domain metrics further clarify that errors in these extreme categories are driven by a small set of outlets. Unlike the diverse 'Slightly Right' group, the 'Far Left' and 'Far Right' categories contain fewer unique domains e.g., *Dom: 1/1* or *Dom: 5/7*, yet these specific outlets consistently trigger model errors. The 'Slightly Left' matrix confirms the specific discrediting pattern discussed in the main manuscript, where models tend to underestimate the reliability of slightly left-leaning sources. A substantial cluster of misclassifications appears in the lower triangle, below the diagonal. This visually demonstrates that models penalize these articles, assigning them lower reliability labels compared to the ground truth, a behavior consistent across all four architectures.



(a) Training and Validation loss over 20 epochs for Llama 3. The green marker indicates the optimal model checkpoint at epoch 4. Longer training durations result in poorer generalisation performance.



(b) Training loss over three epochs for Mistral, shown across five folds.

4 Convergence Analysis

To ascertain the most suitable training duration, a preliminary experiment involving 20 epochs was conducted using the Llama 3 model. As can be seen in Figure 4a, the model converges quickly. The validation loss reaches its minimum at epoch 4 (0.4416), after which the generalisation gap widens, indicating overfitting. The empirical evidence gathered from our most complex architecture validates our decision to limit fine-tuning to five epochs across all models, with the exception noted below.

We conducted an analogous convergence check for Mistral across the five-fold cross-validation. As can be seen in Figure 4b, the training loss decreased sharply from an average of 2.02 at Epoch 1 to 0.36 at Epoch 2 and then to 0.23 at Epoch 3. The average improvement between epochs 2 and 3 is 0.13, indicating diminishing returns and stabilisation. This pattern mirrors the rapid convergence observed for Llama 3, supporting the idea of limiting Mistral training to three epochs to improve efficiency without compromising on convergence.